

1. Title

Expert Decision Replay Platform

2. Objective

Develop a centralized platform that records important organizational decisions, including the problem statement, available alternatives, evaluation criteria, risks, stakeholders involved, discussions, approvals, implementation status, and final outcomes.

The system enables organizations to preserve institutional knowledge, allowing employees to review previous decisions, understand why certain approaches were chosen, and avoid repeating past mistakes.

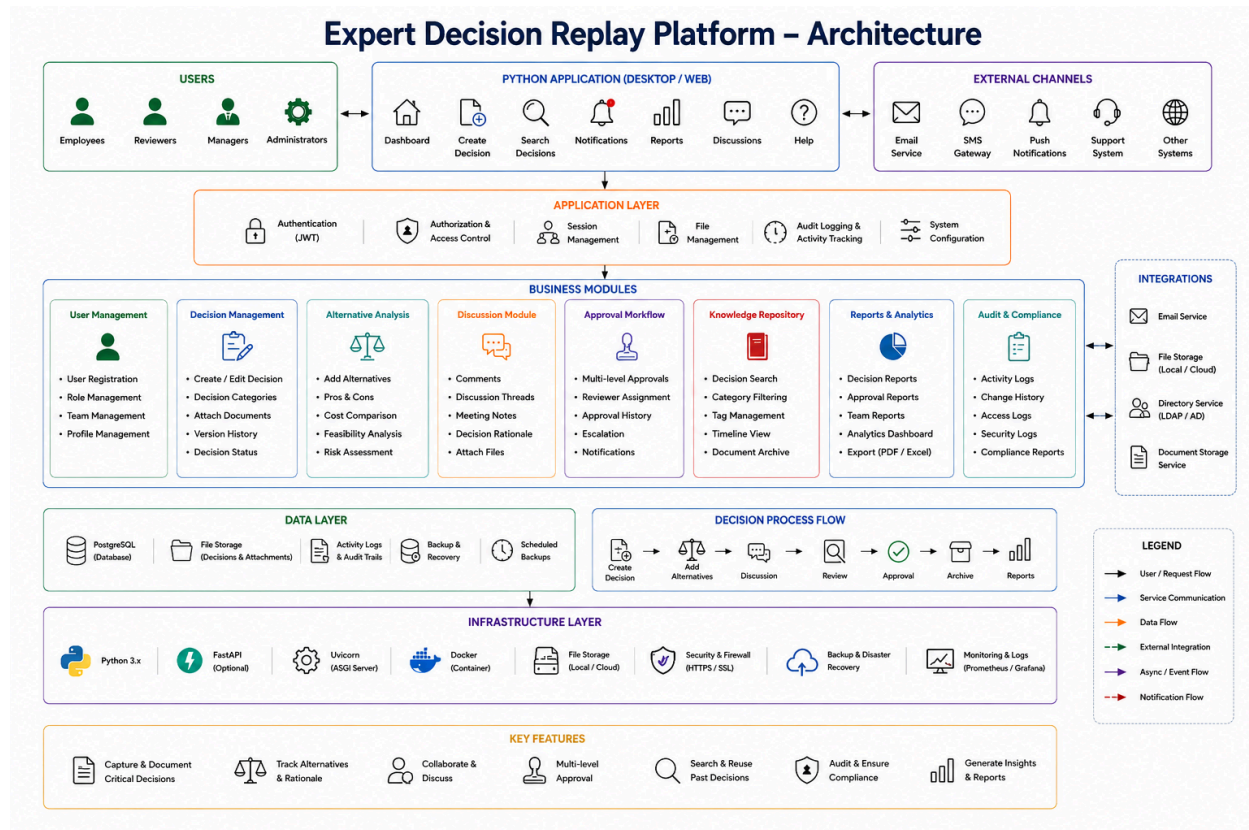
The platform supports structured decision documentation, approval workflows, document management, discussion threads, version history, audit logs, reporting, and decision analytics.

This solution can be used by enterprises, consulting firms, engineering organizations, healthcare institutions, government departments, universities, and research organizations.

Outcomes

- (i) Build a centralized decision management platform.
 - (ii) Implement secure authentication and role-based access.
 - (iii) Develop structured decision creation workflows.
 - (iv) Implement a multi-level approval process.
 - (v) Build discussion and collaboration modules.
 - (vi) Maintain complete decision history and audit logs.
 - (vii) Generate reports and organizational decision analytics.
 - (viii) Deploy using Docker and cloud infrastructure.
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3. Architecture Components



4. Modules to be Implemented

1. User Management

- Registration/Login
- Role Management
- Team Management
- User Profiles

Roles

- Employee
- Reviewer
- Manager
- Administrator

2. Decision Management

- Create decision
- Edit decision
- Decision categories
- Attach documents
- Version history
- Decision status

Decision Status

- Draft
 - Under Review
 - Approved
 - Rejected
 - Archived
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3. Alternative Analysis

- Record multiple options
 - Pros & Cons
 - Cost comparison
 - Feasibility analysis
 - Risk assessment
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4. Discussion Module

- Comments
 - Discussion threads
 - Meeting notes
 - Decision rationale
 - Attach supporting files
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5. Approval Workflow

- Multi-level approvals
- Reviewer assignment

- Approval history
 - Notifications
 - Escalation
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6. Knowledge Repository

- Decision search
 - Category filtering
 - Tag management
 - Timeline view
 - Document archive
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7. Dashboard

Employee Dashboard

- My Decisions
- Pending Reviews
- Recent Activities

Manager Dashboard

- Team Decisions
- Pending Approvals
- Decision Statistics

Admin Dashboard

- System Analytics
 - User Activity
 - Organization Reports
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8. Audit & Compliance

- Activity logs
- Version tracking
- Change history
- Security logs

- Access logs
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9. Reports & Export

- Decision Reports
 - Approval Reports
 - Team Reports
 - Audit Reports
 - PDF Export
 - Excel Export
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5. Week-wise Module Implementation

Milestone 1 (Week 1-2)

Tasks

- Requirement analysis
- Database design
- UI wireframes
- FastAPI setup
- React setup
- Authentication
- User Management

Outcomes

- Project initialized
 - Authentication working
 - Database designed
 - User roles implemented
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Milestone 2 (Week 3-4)

Tasks

- Decision management
- Alternative comparison
- File uploads
- Discussion module
- Version tracking

Outcomes

- Complete decision workflows
 - Collaboration features
 - Document management
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Milestone 3 (Week 5-6)

Tasks

- Approval workflows
- Notifications
- Audit logging
- Reports
- Dashboard

Outcomes

- Approval system completed
 - Reports generated
 - Dashboards functional
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Milestone 4 (Week 7-8)

Tasks

- Testing
- Bug fixing
- Docker deployment
- Documentation
- Final presentation

Outcomes

- Production-ready platform
 - Successful deployment
 - Complete documentation
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6. Evaluation Criteria

Week 2

- Authentication completed
- User management functional
- Database finalized

Week 4

- Decision management completed
- Discussion module working
- Alternative tracking functional

Week 6

- Approval workflows completed
- Reports generated
- Dashboards operational

Week 8

- Deployment completed
 - Testing completed
 - Documentation submitted
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7. Tools & Tech Stack

Backend

- Python
- FastAPI

Frontend

- Tkinter
- PyQt
- Flask

Database

- PostgreSQL
- Redis

Authentication

- JWT
- OAuth2

Backend Libraries

- SQLAlchemy
- Alembic
- Pydantic
- Uvicorn

Storage

- AWS S3 / Local Storage

DevOps

- Docker
- GitHub
- GitHub Actions
- Postman

8. Performance Metrics

Decision Management

- Decision creation rate
- Approval turnaround time
- Search response time

System Performance

- API latency
- Concurrent users
- Database performance

User Activity

- Active users
- Decisions created
- Approval completion rate